# Richard Lin

U.S. Citizen

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## EDUCATION

**Temple University** *Philadelphia, PA*
*Bachelor of Science in Electrical and Computer Engineering* *Expected Graduation: May 2028*
Minor: Computer Science | Fundamentals of Physics Certificate | GPA: 3.88/4.00 | Dean's List: F24, S25, F25
**Courses:** Classical Control Systems, Signals & Systems, Processor Systems, Engineering Computation IV (GPU Programming)
**Honors & Awards:** STEPS Scholarship (NSF-funded), Provost's Scholarship, IEEE SAC SumoBot, 3rd Place (2025)

## PROFESSIONAL EXPERIENCE

**Undergraduate Research Assistant** *Oct 2025 – Present*
*Temple University – Computer Fusion Lab (CFL)* *Philadelphia, PA*
*Advisor: Dr. Li Bai*

- Administered lab systems and website, handling upgrades, troubleshooting, security, and network reliability
- Built a Python multi-agent system with TechOWL to help people with disabilities find the top 3 assistive devices from natural-language requests, using RAG and LLM reranking on a daily-synced catalog of 400+ devices scraped from NATADS
- Trained an ACT model with LeRobot for assistive robotic-arm pickup and handover; implemented ROS-based C++ motion control and integrated STT/TTS for Mini Pupper and the ROBOTIS OP3 humanoid

**Undergraduate Teaching Assistant** *Aug 2025 – Present*
*Temple University* *Philadelphia, PA*

- Assist faculty across three engineering courses with instruction, in class activities, and grading; hold office hours
- **Intro to Engineering:** Guide students in embedded Python and circuit design for Micro:bit incubators; help debug code and wiring, test components (e.g., LEDs, relays, TMP36), and refine temperature control and accessible feedback
- **Engineering Problem Solving:** Help students with Excel and Python (NumPy, Matplotlib) and a wind turbine installation project analyzing TMY weather data (8,760 hours) to compare energy output and breakeven across turbine heights
- **Engineering Computation I:** Teach Linux/Bash, C/C++/Python, time complexity, data structures, and memory management

**STEPS Ambassador** *Sep 2024 – Present*
*Temple University – Sustainable Temple Energy and Power Scholars (STEPS)* *Philadelphia, PA*

- Mentored STEPS underclassmen, and led renewable-energy STEM demos during high school tours and open house

## PROJECTS

**Musical Light Show – ECE 2352: Circuits and Electronics II** *Apr 2026 – May 2026*

- Designed and built an analog audio processing system that combines left/right audio channels and separates high and low frequency signals to drive two audio-reactive LEDs
- Implemented two op-amp stages for summing and volume/gain control (13–100%) using a feedback potentiometer, 219 Hz high/low-pass RC filters, peak detectors, and BJT LED drivers current limited to 30 mA
- Calculated component values using Ohm's Law and KVL; verified Multisim (SPICE) and hardware within 1% of calculations

**Detective Conan AI Glasses (Childhood Dream Project) – In Progress** *Aug 2026 – Present*

## CLUBS & LEADERSHIP

**President** *Sep 2024 – Present*
*Institute of Electrical and Electronics Engineers (IEEE) – Temple University* *Philadelphia, PA*

- Led 14-member IEEE executive board in planning outreach programs, technical workshops, and industry networking events, while coordinating with four affiliated chapters and societies (WIE, PES, EMBS, HKN)
- Expanded IEEE to weekly workshops and major events, boosting turnout from 10 to 20–30+ and SAC participation by 50%
- Founded the first annual multi-university hardware hackathon with Drexel and UPenn IEEE branches

**Vice President** *Jun 2026 – Present*
*Eta Kappa Nu (IEEE-HKN), Iota Sigma Chapter – Temple University* *Philadelphia, PA*

## SKILLS

**Programming:** Python, C, C++, Java, MATLAB, Bash, SystemVerilog, JavaScript, HTML/CSS, CUDA, AVR Assembly
**AI & Robotics:** PyTorch, LLM Integration, RAG, Multi-agent Systems, LeRobot (ACT), OpenCV, ROS, STT/TTS Integration
**Hardware & Tools:** Git/GitHub, Linux, Raspberry Pi, Arduino, FPGA, MultiSim, Soldering, Multimeter, Oscilloscope, Simulink
**Languages:** English, Mandarin Chinese, Fuzhounese, Spanish